

Unit II: Companies Act, 2013 – Part II (30 Questions)

Directors and KMP – Composition of Board and Committees – Board and Committee meetings – Secretarial standards on Board and General Meetings – Management and administration including general meetings and postal ballot – rights of shareholders including right to inspect documents

Unit III: Companies Act, 2013 – Part III (20 Questions)

Loans, Investments, Guarantees / Security – related party transactions – Borrowings

Unit IV: Corporate Restructuring and Dissolution (20 Questions)

Mergers and amalgamations – strike-off – vanishing companies – willful defaulter – winding up – valuation of shares and business – matters relating to joint venture agreements – shareholder agreements, including share subscription and share purchase agreements – due diligence exercises – arbitration clauses – non- is closure agreements

Unit V: Capital Markets and Securities Laws (20 Questions)

SEBI (Listing Obligations and Disclosure Requirements) Regulations, 2015, including compliance relating to related party transactions with knowledge on AS-18/Ind-AS 24 - SEBI (Substantial Acquisition of Shares and Takeover) Regulations, 2011 – SEBI (Prohibition of Insider Trading) Regulations, 2015 – SEBI (Issue of Capital and Disclosure Requirements) Regulations, 2018 – Basics of primary and secondary market, including compliance with stock exchanges

Unit VI: Other Laws – Part I (20 Questions)

Indian Contract Act, 1872 – Negotiable Instruments Act, 1881 – Specific Relief Act, 1963 – Arbitration and Conciliation Act, 1996 – Laws relating to intellectual property rights

Unit VII: Other Laws – Part II (30 Questions)

Transfer of Property Act, 1882 – Right to Information Act, 2005 – Tamil Nadu Transparency in Tenders Act, 1998 – Information Technology Act, 2000 – Prevention of Money Laundering Act, 2002 – Foreign Exchange Management Act, 1999 – Fugitive Economic Offender Act, 2018 – Environmental Laws – Sexual Harassment of Women at Workplace (Prevention, Prohibition and Redressal) Act, 2013 – Competition Act, 2002

Unit VIII: ESG and BRSR (10 Questions)

Sustainable development – ESG reporting – Business Responsibility and Sustainability Report – Social Stock Exchanges – Green Bonds – Carbon Rating / Green Rating.

Unit IX: Drafting and Compliance Management (10 Questions)

Drafting of resolutions, agreements etc. – compliance management – audit and due diligence – secretarial audit – forensic audit – fraud detection techniques.

Unit X: Strategic Management (10 Questions)

Strategic analysis – formulation of strategy – sources of corporate funding – project evaluation.

**4. Basics of Engineering
(Degree Standard)****Code: 422****UNIT I: Mathematics (20 Questions)**

Matrices: Eigen values – Eigen vectors of a real matrix - Cayley - Hamilton theorem - Similar and Orthogonal transformations - Reduction of a quadratic form to Canonical form by orthogonal transformation. Ordinary differential equations: Order and degree - Higher order linear ODE with constant coefficients - Method of undetermined coefficients - Method of variation of parameters - Cauchy's and Legendre's linear equations. Functions of several variables: Partial derivatives - Total derivatives - Euler's theorem - Implicit functions - Jacobians - Taylor's theorem - Maxima and Minima. Integration: Definite and indefinite Integrals - Techniques of integration using integration by parts and Trigonometric Integrals - Double Integrals - Change of order of integration - Volume Integrals. Vector Calculus: Vectors and

scalars - Gradient and Directional derivatives - Divergence and Curl - Applications of Green's theorem, Gauss divergence theorem and Stoke's theorem. Complex variables: Analytic functions - Verification of Analyticity - Construction of Analytic functions - Conformal Mappings - Bilinear transformations. Complex Integration: Cauchy's integral theorem - Cauchy's fundamental theorem - Cauchy's residue theorem - Taylor's and Laurent's series - Contour integration (excluding poles on the real axis). Laplace transform: Existence of Laplace transform - Laplace transform of elementary functions- Properties - Laplace transform of Periodic functions - Inverse Laplace transform - Convolution theorem - Solution of linear second order ODE by Laplace transform technique.

UNIT II: Engineering Physics (25 Questions)

Mechanics: Newton's laws of motion – gravitation – work, energy and power - Properties of matter : Elasticity – moduli of elasticity - Sound : intensity level – reverberation – Ultrasonics : production, detection and applications - Thermal Physics : Thermal expansion - thermal stress - expansion joints - bimetallic strips - thermal conductivity- heat conductions in solids – flow of heat through compound media – Thermodynamics – Laws of thermodynamics – Carnot engine - Applied Optics : Interference – Young's double slit experiment - anti-reflection coatings - Diffraction - Lasers – principle and applications – CO₂ and Nd:YAG laser - semiconductor lasers – applications of Lasers – Optical fibres: classification (index & mode based) - principle and propagation of light in optical fibres - acceptance angle and numerical aperture - fibre optic communication system - Quantum Physics : Photoelectric effect– dual nature of matter and radiation – Heisenberg's uncertainty principle - Schrödinger's wave equation - Physics of Materials : Crystal structures – unit cell – packing factor – Superconductivity : Properties and applications - Magnetisation of matter: Magnetic dipole moment – atomic magnetic moments- magnetic permeability and susceptibility - Magnetic material classification : diamagnetism – paramagnetism – ferromagnetism – Semiconductors : Intrinsic Semiconductors – Energy band diagram – direct and indirect band gap - extrinsic semiconductors – Dielectric materials: Matter polarization and relative permittivity - dipole moment and polarization vector -polarization mechanisms: electronic, ionic, orientational, interfacial and total polarization- frequency dependence - dielectric strength and break-down in gases, liquids and solids.

UNIT III: Engineering Chemistry (25 Questions)

Fuel -Classification of fuels - Calorific value - Solid fuel - Liquid fuel - Gaseous fuel - Octane number - Cetane Number -Lubricants - Classification - Greases - Solid Lubricants. Water - Sources - Classifications - Softening process - Desalination - RO Method - Internal treatment - Treatment of Water for Municipal purposes. Plastics - High polymer - classification - Polymerization techniques - Thermoplastics - Thermosetting resins - examples. Rubber -"Types of Rubber - Vulcanisation - Properties- Unvulcanised and Vulcanised. Natural Rubber - Synthetic Rubber - examples. Refractories - Classification - Manufacture of Refractories - Magnesite - Silica - Zirconia -Chromite. Abrasives - Natural - Artificial-Abrasive paper & cloth. Corrosion: Dry and Wet corrosion - Factors affecting corrosion-Different types of corrosion. Productive coating - Hot dipping- metal cladding, electro deposition - Organic Coatings - Paints - Varnishes. Cement and lime- setting and hardening. Explosives- classifications - characteristics-requirements for good explosives- nitrocellulose- TNT- TNB-DNB-PETN-RDX. Alloys- purpose of making alloy- types of alloys- Ferrous alloys. Electrochemistry - conductors and non-conductors - Kohlrausch law - Electrochemical cell- reversible and irreversible cells - EMF - Concentration cell- polarization - over voltage, decomposition potential. Fuel Cells. Nano Chemistry-Basics- distinction between molecules, Nano materials and bulk materials. Size dependent properties and applications of Nano Materials.

UNIT IV: Basics of Computer Engineering (25 Questions)

Computer Organisation - CPU and Microprocessor [ALU, Control Unit and Bus Structure] - Data Storage [Primary, Secondary and Virtual] - Input and Output Devices. System Software - Assembler - Compiler - Loader - Linker - Operating Systems. Programming Languages - Classification of Programming Language, Algorithm, Flow chart, Pseudo code, High-Level Languages – Fundamental concepts of C Programming. Basic Computer Networking - Network Components [Routers, Bridges, Gateways] - ISO-OSI Reference Model - LAN - WAN - Client-Server Architecture - Internet - World Wide Web. Applications - Office Tools - Word processor - Spreadsheet - Power point - Introduction to Database concepts - E-mail - Browser. IT Enabled Services - E-Governance - E-Commerce - Multimedia.

UNIT V: Basics of Civil and Mechanical Engineering (30 Questions)

Introduction to Engineering mechanics - Units and Dimensions - Laws of Mechanics - Coplanar Forces - Static Equilibrium of Rigid body - Moment of force - free body diagram - friction - laws of friction - sliding friction - wedge friction - Rolling resistance - Ladder friction - Friction in screws - Screw jack -Belt friction - Properties of surfaces and solids - Centroids and centre of mass - line and areas - Rectangular, circular, triangular areas by integration - T-section, I- Section, Angle section, Hollow section - Area moment of inertia of plane areas - Parallel axis theorem – Perpendicular axis theorem, Polar moment of Inertia, Principle moment of Inertia Mass moment of inertia- Centroid of the simple solids - Dynamics of particle - Displacement, velocity and acceleration - Different types of motion - Rectilinear , Curvilinear and Projectile motions - Newton's II-law of motion - Work Energy equation - Impulse and momentum principles.

UNIT VI: Basics of Electrical and Electronics Engineering (30 Questions)

Ohm's law – Kirchoff's laws – Introduction to DC and AC circuits - single phase and three phase circuits – Power and Power factor, Unbalanced and Balanced loads, Operating principles of moving coil and moving iron instruments (voltmeters and ammeters) – wattmeters, multimeter, energy meters and megger, Construction and principle of operation: DC motors- DC generators-Transformers- Induction motors, Characteristics of PN junction diode - zener diode- half wave and full wave rectifiers - Bipolar junction transistor (CC,CE,CB configurations), SCR, Amplifiers- Operational amplifiers – Inverting and Non-inverting amplifiers, Binary number system- logic gates- Boolean algebra - Half and full adders- Flip-flops -registers and counters- A/D and D/A conversion, Types of analog and digital signals- Modulation and Demodulation(amplitude and frequency) Communication systems: Radio- TV- Fax- Microwave-Satellite and optical fibre.

UNIT VII: Principles of Management (15 Questions)

Management - Definition, Evolution of Management Philosophies, Types of Business, Environment Analysis - Planning- Types, Steps, Forecasting, MBO, MBE. Organizing – Departmentation, Line and Staff Authority, Delegation and Decentralization. Staffing - Manpower Planning, Recruitment and Selection, Training, Performance Appraisal. Directing – Theories of Motivation, Leadership Styles, Power and Politics, Change Management, Conflict Management, Communication in Business- Controlling Types, Control Techniques, Budgetary and Non-Budgetary Control.

UNIT VIII: Total Quality Management (15 Questions)

Quality – Definitions, Vision, Mission and Policy statements-Dimensions of Product and Service Quality-Contributions of Quality Gurus-Deming, Juran, Crosby, Masaaki Imai, Feigenbaum, Ishikawa. Costs of Quality- Continuous Process Improvement- PDCA, Quality Circle, 5S, Kaizen-Statistical Process Control (SPC), 7QC Tools, New Management Tools of Quality, Bench Marking, 6 sigma, Quality Function Deployment (QFD), POKAYOKE, Total Productive Maintenance (TPM), Business Process Reengineering (BPR), Quality Certifications.

UNIT IX: Environmental Science and Engineering (15 Questions)

Definition, scope and importance of environment – need for public awareness. Eco-system and Energy flow– ecological succession. Types of biodiversity: genetic, species and ecosystem diversity– values of biodiversity, India as a mega-diversity nation – hot-spots of biodiversity – threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts – endangered and endemic species of India – conservation of biodiversity: In-situ and ex-situ. Environmental pollution: Causes, Effects and Preventive measures of Water, Soil, Air and Noise Pollutions. Solid, Hazardous and E-Waste management. Energy management and conservation, New Energy Sources - Need of new sources. Different types of new energy sources. Applications of- Hydrogen energy, Ocean energy resources, Tidal energy conversion. Concept, origin and power plants of geothermal energy. Sustainability and management - Development , GDP, Sustainability- concept, needs and challenges-economic, social and aspects of sustainability-from unsustainability to sustainability-millennium development goals, and protocols-Sustainable Development Goals-targets, indicators and intervention areas. Climate change- Global, Regional and local environmental issues and possible solutions. Concept of Carbon Credit - Carbon Footprint. Environmental management in industry- Material Life cycle assessment, Environmental Impact Assessment. Sustainable habitat: Green buildings, Green materials, Energy efficiency, Sustainable transports. Sustainable energy: Non-conventional Sources, Energy Cycles carbon cycle, emission and

sequestration, Green Engineering: Sustainable urbanization- Socio-economical and technological change.

5. Business Administration, Financial and Cost Accountancy (PG Degree Standard)

Code: 525

Unit I: Accounting (20 Questions)

- Accounting for non - profit organizations and incomplete records
- Branch and Departmental Accounts (including foreign branches)
- Accounting Standards (AS) – Generally Accepted Accounting Principles (GAAP) – AS - applicability, interpretation, scope and compliance – International Financial Reporting Standards (IFRS) - Ind AS 2, 10, 115, 16, 21, 20, 23, 37 & 38.
- Company Accounts - Presentation of Financial Statements as per Schedule III of Companies Act 2013 – Statement of Profit or Loss, Balance Sheet, Change in Equity.
- Accounting of Banking

Unit II: Auditing (15 Questions)

- Nature and scope and significance of auditing
- Audit Engagement, Audit Programme, Audit working papers, Audit note book audit evidence and audit report
- Internal check, internal control and internal audit - industry specific
- Reporting requirements of under companies act – report vs certificate, contents of the report and qualifications in the report.
- Cost audit and Secretarial audit
- Auditing of different types of undertaking - education, hospitals, cooperative societies, trusts.

Unit III: Financial Management and Business Evaluation (25 Questions)

- Meaning – objectives - Scope of Financial Management
- Sources of Funds - Introduction of Financial Markets - Capital & Money Markets
- Tools for Financial Analysis - Ratio, Funds flow analysis, Cash flow analysis
- Cost of Capital, Capital Structure and Leverages
- Working Capital Management – Financing - Cash, Inventory, Receivables and Payables
- Capital Budgeting - Valuation Methods - Non Discounted cash flow method-Discounted Cash Flow Method - Free Cash flow Method, Net Present Value Method (NPV) - Profitability Index Method (PI) - Internal Rate of Return Method (IRR) -Valuation of Inventory - Valuation of Investments - Valuation of Intangibles - Valuation of Human Resources - Valuation of Equity
- Risk Management - risk measurement - risk analysis - Value at Risk (VAR) - Capital Adequacy Norms & Income Recognition Norms (NPA) in NBFCs.

Unit IV: Taxation (20 Questions)

- Direct Taxation - Residential Status - Heads of Income - assessment of income of different persons – Tax Deducted at Source (TDS) / Tax Collected at Source (TCS) / Advance Tax - Income Computation and Disclosure standards (Basic Concepts only) - Taxation as applicable to Government undertaking and Cooperative Societies. Due dates and compliances. Effects of non-compliance and penalty. Tax audit provisions under Income Tax Act, 1961.
- Indirect Taxation - Canons of Taxation - Customs Law - Goods and Services Tax (GST) – Central GST (CGST) – State GST (SGST) – Integrated GST (IGST) - Cess - Registration – Supply - Time of Supply - Value of Supply - Place of Supply - E-Invoicing E-Way Bill, Input tax credit - Reverse Charge Mechanism (RCM) - Payment of Taxes, Penalties.

UNIT V: Laws and Regulations (25 Questions)

- Micro, Small and Medium Enterprises Development (MSMED) Act 2006.
- Factories Act 1948.